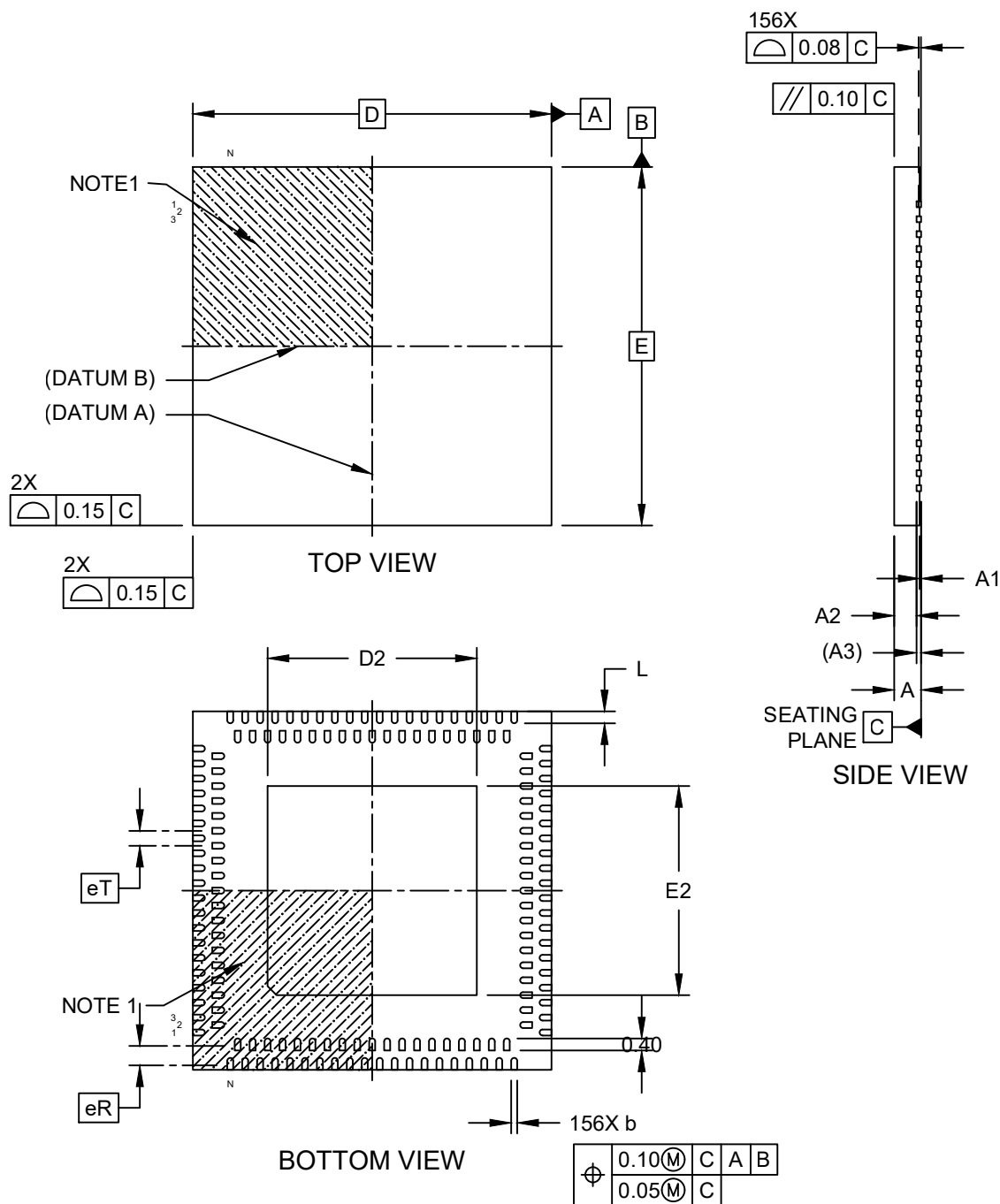


Packaging Diagrams and Parameters

156-Lead Very-Thin Quad Flatpack No-Leads (8NW) Dual Ring - 12x12x0.9mm Body [VQFN] With 7x7mm Exposed Pad

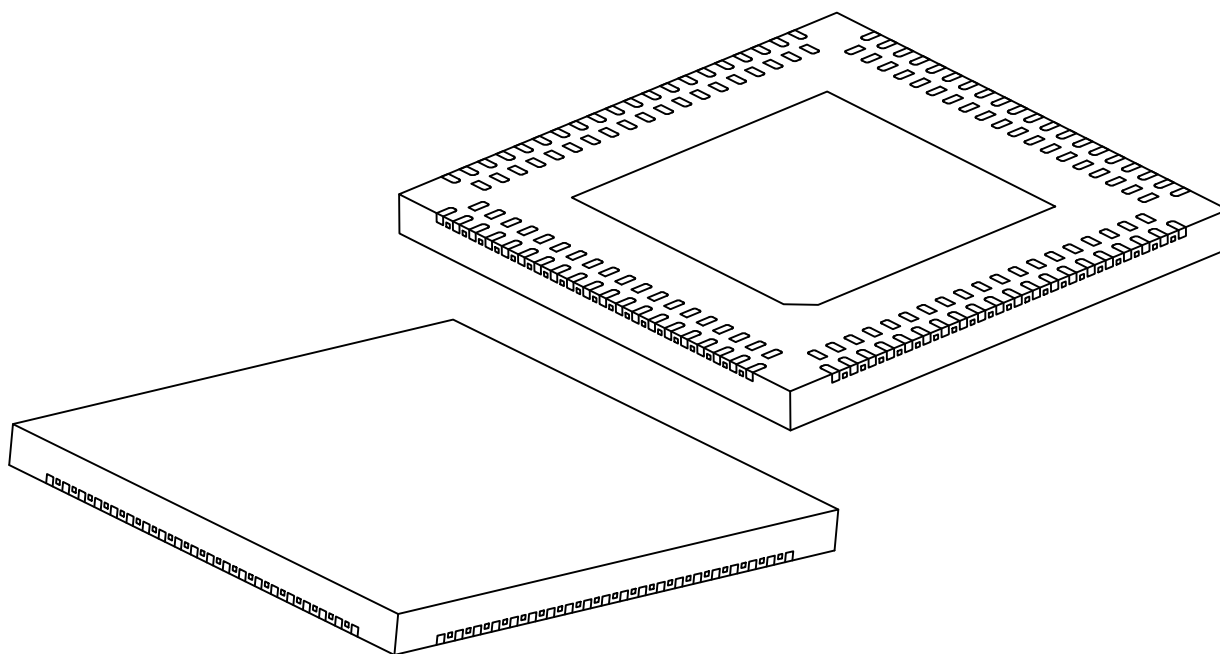
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

156-Lead Very-Thin Quad Flatpack No-Leads (8NW) Dual Ring - 12x12x0.9mm Body [VQFN] With 7x7mm Exposed Pad

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		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		156		
Pitch of Lead Fingers	eT		0.50 BSC		
Pitch of Lead Fingers Rows	eR		0.65 BSC		
Overall Height	A		0.80	0.85	0.90
Standoff	A1		0.00	0.02	0.05
Mold Thickness	A2		0.65	0.70	0.75
Terminal Thickness	A3		0.152 REF		
Overall Length	D		12.00 BSC		
Exposed Pad Length	D2		6.90	7.00	7.10
Overall Width	E		12.00 BSC		
Exposed Pad Width	E2		6.90	7.00	7.10
Terminal Width	b		0.16	0.20	0.28
Terminal Length	L		0.30	0.40	0.50
Terminal-to-Exposed-Pad	K		1.45 REF		

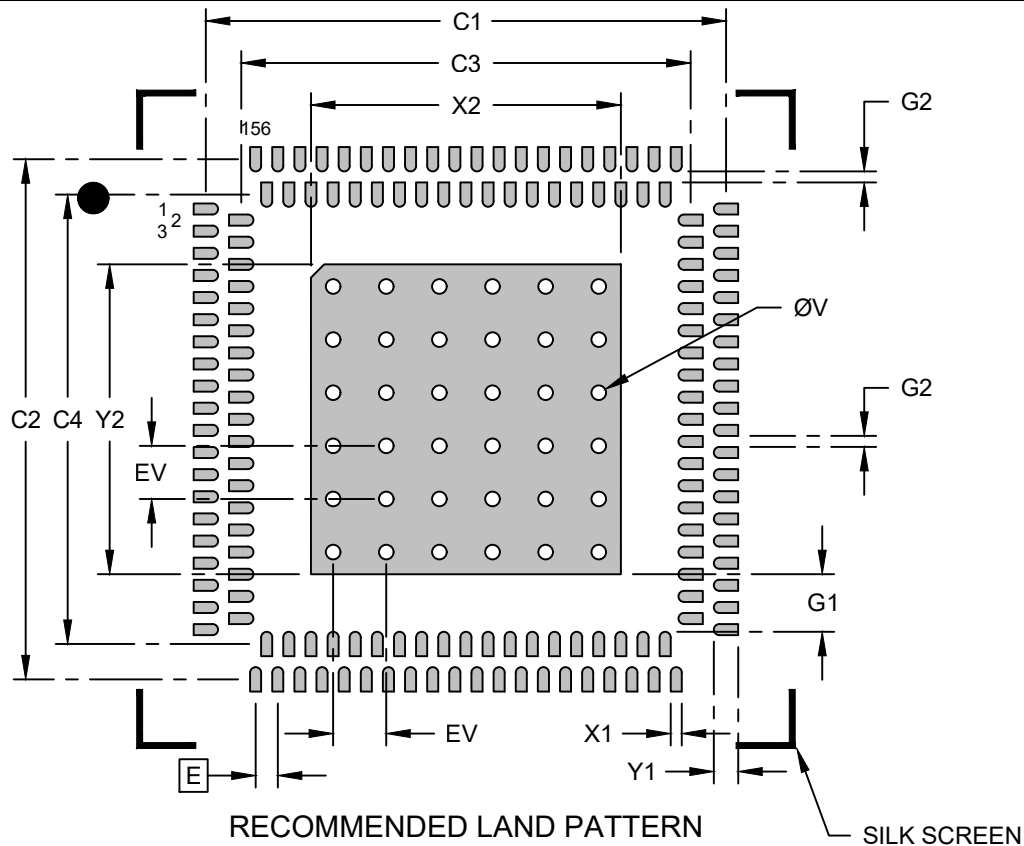
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerances, for information purposes only.

Land Pattern

156-Lead Very-Thin Quad Flatpack No-Leads (8NW) Dual Ring - 12x12x0.9mm Body [VQFN] With 7x7mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Center Pad Width	X2			7.70
Center Pad Length	Y2			7.70
Contact Pad Spacing (Outer Row)	C1		11.75	
Contact Pad Spacing (Outer Row)	C2		11.75	
Contact Pad Spacing (Inner Row)	C3		10.15	
Contact Pad Spacing (Inner Row)	C4		10.15	
Contact Pad Width (X156)	X1			0.25
Contact Pad Length (X156)	Y1			0.55
Contact Pad to Center Pad (X76)	G1	1.30		
Contact Pad to Contact Pad (X148)	G2	0.25		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, please refer to current industry standard IPC-7093.

Microchip Technology Drawing C04-02667 Rev A